

Adjudication of Three Independent Audits of the Prime-Spectral Framework Manuscript

Working review document

October 5, 2026

1 Scope, material, and method

Three independent line-level audits of the manuscript *The Prime-Spectral Framework: Theorems, Conjectures, and Numerical Tests* were received in `audits/prime_audit.txt`: **astra** (13 points, short-form), **grok** (13 numbered flaws, a 7-row internal-inconsistency table, and a 7-item improvement program), and **muse** (8 subsections of mathematical gaps, minor issues, and a 6-item improvement plan). This document adjudicates them jointly: every substantive claim was checked against the manuscript text, every checkable mathematical assertion was re-derived or computed, the numerical demands that could be executed at laptop scale were executed, the surviving fixes were folded into the manuscript, and the points on which the audits disagree with each other were resolved on the mathematics.

Verdict categories. Each point receives one of: **applied** (the claim is correct and a fix has been applied); **partial** (the concern is valid but the stated specifics are wrong or overstated; a precise fix is applied instead); **rebutted** (the claim is factually wrong, with proof or computation); **already satisfied** (the manuscript already contains the demanded content; the audit misread or reviewed an extraction artifact); **run** (the demand was executed as a new computation and folded in).

Verification tools. (i) Exact and Monte-Carlo checks (`scripts/adjudication_checks.py`): counting-defect convergence, conjugacy-class divisibility on 12 finite groups, the Dirichlet return-time bound, and Erdős–Kac normal-order simulation. (ii) New numerical controls (`c4_scaled_eth.py --scramble, --sigma-quantile`): label-scrambled diagonal controls, a generic-coupling quantile sweep, and $1/\log D$ scaling fits. (iii) Line-level re-reading of all eight source files of the manuscript.

2 Cross-audit oppositions and their resolution

The audits disagree with each other on nine points of substance. Each opposition is stated, adjudicated, and cross-referenced to the change made.

Opposition	Resolution	Manuscript change
O1. Counting-law defect: grok — “non-convergent sawtooth of amplitude $\log 2$ ”; muse — “error is $O(e^{-E/\hbar\omega_0})$ ”; the audited text said “ $O(1)$, does not converge.”	<i>Muse is right.</i> Computed: $r(E) = \log[e^x] - x$ satisfies $\log(1 - e^{-x}) \leq r \leq 0$ with $r \rightarrow 0$ like e^{-x} ; the jump amplitude at level m is $\log(1 + 1/m) \sim 1/m$. The $\log 2$ bound is only the global envelope at the bottom of the spectrum; the non-convergent object is the fractional part $\{e^x\}$, not the error.	Theorem ?? restated: sharp two-sided bound, exponential decay, smoothing item, staircase honesty; C5 fluctuation sector reassigned from the floor to the prime-counting oscillations.
O2. Independence of $\{\log p\}$: grok — “Baker/Lindemann–Weierstrass territory”; muse — “unique factorization suffices, no Baker needed.”	<i>Muse is right.</i> The paper’s proof clears denominators and exponentiates: $\prod p_i^{c_i} = 1$ forces $c_i = 0$. Baker’s theorem concerns algebraic coefficients and is neither needed nor used.	Lemma statement made explicitly finite-support (astra’s variant of the same point).
O3. Hardy–Ramanujan citation: grok — the partition asymptotics $p(n) \sim e^{\pi\sqrt{2n/3}}$ is cited for the cascade (category error); muse — normal order vs. maximal order conflated in prose.	<i>Both charges miss the text.</i> The manuscript cites hardyramanujan1917 — “The normal number of prime factors of a number n ”, QJM 48 (1917) — which <i>is</i> the normal-order theorem; the prose distinguishes typical from maximally composite integers.	Abstract wording strengthened to “Hardy–Ramanujan normal-order and Erdős–Kac theorems” to remove the ambiguity the audits read in.
O4. C9 class sizes: grok — “class sizes divide the group order only in trivial cases”; the protocol fits $c_i \mid N$.	<i>The paper is right.</i> Orbit–stabilizer: $ C(x) \cdot C_G(x) = G $, so $ C \mid G $ always. Brute-force verified on C_2 – C_5 , S_3 – S_5 , D_4 – D_8 , Q_8 : every class size divides $ G $. The real constraints are realizability and analyst degrees of freedom, which grok also raises (see G10).	Register C9 row: $c_i = C_i $ with the orbit–stabilizer gloss; pre-registration and non-confirmatory Z-pole wording added.
O5. Microcanonical temperature: grok — “exactly critical at every energy” is circular and false; muse — no phase transition is proved, rename; astra — interpretive analogy.	<i>All three converge on demotion; grok’s diagnosis of the mechanism is correct.</i> The unsmoothed entropy is a staircase (derivative 0 a.e.); $T \equiv T_H$ holds only for fixed-window smoothings, whose growth rate is $1/\hbar\omega_0$ exactly.	Theorem restated as a growth-rate statement with the staircase made explicit; “exactly critical” removed; Hagedorn reading tied to unit degeneracy and densification.

Opposition	Resolution	Manuscript change
O6. Canonical ensemble at the pole: grok — no ensemble exists at $T \geq T_H$ and no theorem bridges the divergence to the counting; muse — same, plus a cutoff-canonical demand.	<i>Valid; the bridge is now a proposition.</i> For $\sigma > 1$, $Z(\sigma) = \zeta(\sigma) = \int e^{-\sigma x} dN(x)$ exactly (Laplace–Stieltjes identity); the pole $1/(\sigma - 1)$ is exactly the Laplace transform of the exponential counting law, and $\zeta - 1/(\sigma - 1)$ is entire.	New Proposition (ensemble bridge) in §2; canonical statements confined to $T < T_H$; above- T_H statements labeled microcanonical or cutoff.
O7. Holographic error budget: grok — area-independent $O(1)$ cannot absorb $\log(A/\ell_P^2)$, so Theorem 6.2 is false under the register’s own risk note; astra — the cutoff is an assumption.	<i>grok’s arithmetic is right and is adopted as structure.</i> The counting contributes no $\log A$ term (its defect is exponentially small); every logarithmic correction must be carried by the dictionary.	Conjecture restated with the dictionary slot $s(A) = O(\log(A/\ell_P^2))$; conditional theorem: area coefficient exact, corrections carried by s ; the pure ($s = O(1)$) sub-hypothesis is falsifiable by computed $-\frac{3}{2} \log A$ corrections.
O8. C4 numerics: grok — GOE on a sparse perturbation of a separable ramp is the default expectation; demands a label-scrambled control, a generic quantile sweep, and $1/\log D$ scaling; muse — $D \lesssim 6 \times 10^4$ is pre-asymptotic ($\log \log D$ small).	<i>Both executed.</i> The scrambled controls <i>confirm</i> grok’s reading for level statistics (matched-dose $\langle r \rangle$ is scramble-insensitive: $0.518 \rightarrow 0.512$, $0.509 \rightarrow 0.502$) and <i>refute</i> it for the generic family’s delocalization (scrambling collapses PR/ D by $2\times$ at $d = 4$ and $12\times$ at $d = 3$, and inflates σ_{ETH} $1.8 \rightarrow 4.5$): the arithmetic diagonal is a transport organizer, not a source of level repulsion. muse’s asymptotic-distance point is quantified ($\log \log$ of the largest truncation ≈ 4.5).	Remark (f) label-scrambled controls and (g) finite-size scaling added; Figure ??; register rows C4/C5 updated; the scaling limit is reported as unsettled.
O9. Editorial strategy: grok — split the paper into a mathematical note and a conjectural essay; muse — quarantine the interpretation inside one paper; astra — enforce a per-claim ledger.	<i>muse and astra adopted.</i> The register <i>is</i> the per-claim ledger (statement, grade, dependencies, falsifier, exemption markers); the two-plane presentation quarantines interpretive dependencies; splitting is a publishing decision without mathematical content, noted here for the record.	No structural split; §1.2 amended (operational testability for C1–C11, interpretive exemption marked); “not implied” boundary paragraph added at the end of §2.

3 Point-by-point verdicts

3.1 astra

Claim	Verdict	Action
A1 Formal construction vs. physical interpretation must be separated (ζ as partition function vs. Hagedorn reading).	partial	Hagedorn reading tied to unit degeneracy/densification; Gibbs existence confined to $T < T_H$; ensemble bridge added.
A2 $N(E)$ needs explicit definition, vacuum convention, proof.	already satisfied	Already exact ($n \geq 1$, $N(E) = \lfloor e^{E/h\omega_0} \rfloor$, vacuum $ 1\rangle$ at $E = 0$); now strengthened with the sharp defect.
A3 “Naive infinite tensor product is the wrong home” is too broad; incomplete tensor products are separable.	applied	Incomplete tensor product over the vacuum Ω identified, unitary to bosonic Fock; the full product never used.
A4 $\mathbb{Q}$ -independence of $\{\log p\}$ needs the finite-support condition.	applied	Lemma restated for finite relations.
A5 Gaussian-prime/chirality language conflates splitting classification with chiral gauge representation.	applied	Factorization supplies no dimension count and no chirality operator; both are marked dictionary entries.
A6 Galois unification under-specified (profinite, not Lie; no functor).	applied	Minimal well-posed dictionary (conj:gut) with the “does and does not supply” boundary; no functor claimed.
A7 Langlands invoked too imprecisely.	already satisfied	Exact setting given (function-field GL_n , [?] in the manuscript); specialization requirement stated.
A8 “Photon as reciprocity map” not well typed.	applied	Dictionary rows labeled demoted; new bookkeeping paragraph: no gauge field, connection, action, or propagating degree of freedom is constructed.
A9 Chebotarev statistics do not establish confinement or color.	applied	Same paragraph; confinement observables explicitly not derived.
A10 “Decidable” overstates.	applied	Replaced by “operationally testable”; C1–C11 carry routes; interpretive entries exempt and marked.
A11 Numerical standards need reproducibility content (seeds, grids, cutoff dependence).	partial	“Code, seeds, parameter grids, and all result files accompany the paper (repository)”; window/cutoff dependence is reported in Remark (a),(f),(g); Monte-Carlo false-positive calibration prices the C9 multiple-testing.

Claim	Verdict	Action
A12 “Gravity as entropic $1/N$ effect” requires definitions.	already satisfied	The claim is register-held (C1/C6, grade B/C) with the falsifier column carrying the operational content; no theorem asserts it.
A13 Per-claim ledger; neutral language in theorem sections.	already satisfied	The register is the ledger; theorem environments state mathematics only; physical identifications live in the register.

3.2 grok

Claim	Verdict	Action
G1 Abstract overclaims the counting law (“exact identity”, $O(1)$, non-convergent sawtooth; C5 unfalsifiable).	partial	See O1: defect decays exponentially; abstract restated; C5’s fluctuation sector reassigned to prime-counting oscillations with the von Koch bound.
G2 “Exactly critical at every energy” is circular/false.	applied	See O5; statement now a growth-rate property; pointwise temperature undefined between levels.
G3 Hardy–Ramanujan is the wrong theorem.	rebutted	See O3: the 1917 normal-order paper is cited, not the partition asymptotics; abstract disambiguated anyway.
G4 Canonical $Z = \zeta$ is not the ensemble the physics needs; no microcanonical theorem; no bridge.	applied	See O6: ensemble-bridge proposition; canonical confined below T_H .
G5 Multiplication operators are not interactions; the diagonalized model is not the conjectured one.	partial	“Selected, not derived” provenance remark; the M_n algebra supplies exact energy bookkeeping (new commutator identity) and the graph; bare H_{tot} is excluded from the conjecture explicitly.
G6 Gaussian extension yields no chirality or five dimensions.	applied	See A5; units quotiented; norm-based energies only.
G7 Holographic bridge is an assumption wearing a theorem number; $O(1)$ cannot absorb $\log A$.	applied	See O7: dictionary slot $s(A)$; area coefficient exact; pure form falsifiable.
G8 Clock thermodynamics contradicts “exactly critical”; no directional limit.	applied	Regulator ν separated from inverse temperature; clock exists on both sides of T_H ; new theorem addendum: no limit $\nu \downarrow \frac{1}{2}$ selects a direction.

Claim	Verdict	Action
G9 Phase incommensurability does not imply the universe cannot repeat; C2's falsifier is absorbable.	partial	Ray-periodicity classification added; quantitative return-time proposition; C2 falsifier restated as a scaling criterion (super-polynomial-in- S recurrence is inconsistent with every cutoff).
G10 C9's falsifier cannot do its work (divisibility, analyst freedom, absence-as-confirmation, single-channel, Kronecker-Weber abelian).	partial	Divisibility charge rebutted (O4); pre-registration language, non-confirmatory Z -pole wording, and single-channel caveat applied; KW already scoped to the abelian row.
G11 C4 numerics do not test thermalization of the arithmetic universe (no scrambled control, no generic quantile sweep, dose tuning, structural ETH failure, no $1/\log D$ scaling).	applied/run	All diagnostics executed (O8): matched-dose GOE is graph-generic (honest concession), generic-family delocalization is arithmetic; full generic sweep 0.46–0.540; scaling fits with the limit reported unsettled; matched dose is the controlled variable, stated as such.
G12 Register self-violations (C4 depth placement, C12 exempt, C10/C11 open-ended, cut-off constraint).	applied	Two-plane DAG separates physical ordering from logical autonomy; §1.2 amended; C10 constraint labeled one-sided; C12 marked interpretive with the exemption.
G13 Small consistency errors ($\ell^2(\mathbb{N}, \times)$, k_B mixing, Bochner toeplitz, “three structural requirements met”).	applied	ℓ^2 of $\mathbb{N}_{\geq 1}$; units paragraph ($k_B = 1$ unless displayed, σ dimensionless, ν regulator); kinematical-level disclaimer for the clock section; abstract now says “obstructions are discharged”.

The 7-row internal-inconsistency table and the 7-item improvement program map onto the rows above (rows 1–7 of the table duplicate G4, G8, G7, G12, G2, G3, G10; improvements 1–7 map to O9, O7, O6, O8/G11, G10, G13, and the new “not implied” boundary respectively). grok’s acknowledgment list (the arithmetic core is correct) is consistent with the manuscript’s own grading and requires no action.

3.3 muse

Claim	Verdict	Action
M1 (2.1) Fock isomorphism not proved; notation; Gaussian $\mathbb{Q}_{>0}$ domain unbounded below.	partial	Fock unitary and incomplete tensor product added; notation already single-convention; the unbounded-below charge is a misread (energies are $\log N(\pi)$ over Gaussian-prime norms, never $\log r$ for $r \in \mathbb{Q}_{>0}$).
M2 (2.2) H_P domain, essential self-adjointness, spectrum qualifiers, overselling.	applied	Explicit $D(H_P)$; compact resolvent, pure point, no finite accumulation, gaps $\hbar\omega_0 \log(1 + 1/n)$; the diagonal-operator nature is stated without depth claims.
M3 (2.3) Trace-class condition; Hagedorn naming; cascade is statistical, not dynamical.	applied	$\beta\hbar\omega_0 > 1$ carried through; degeneracy-one structural difference stated; the statics/dynamics dichotomy is the section's spine (frozen-dynamics proposition + Erdős–Kac theorem).
M4 (2.4) Ensemble window undefined; normal vs. maximal order.	partial	Cumulative measure labeled, shell transfer noted (normal order holds on every dyadic range); typical vs. maximally composite already distinguished.
M5 (2.5) Monoid relations, boundedness, lattice definition.	already satisfied	Cuntz-type relations, range projections, shift ladder already in the monoid proposition; Kato–Rellich now explicit; H_W edges = divisibility graph = occupation axes, stated.
M6 (2.6) Periodicity quantifiers; Bohr σ ; quantitative recurrence.	applied	Ray version added; $\sigma = 2\nu > 1$ fixed in the theorem; new quantitative return-time proposition with polynomial-in-entropy arithmetic scale vs. double-exponential generic scale.
M7 (2.7) Entanglement bipartition-relativity.	already satisfied	The section defines entanglement exactly across prime-mode bipartitions; the product-basis theorem is scoped; the counterexamples are the section's own.
M8 (2.8) Dynamics vs. kinematics; $Z \neq \zeta$ under V ; reporting.	applied	Well-posedness proposition (bounded perturbation, Kato–Rellich); free-theorem statements scoped to H_P ; the interacting family is the conjecture's object, tested in Remark (a)–(g).

Claim	Verdict	Action
M9 (§3) Conjecture-level demands (C1 derivation, C3 dictionary, C4 pre-asymptotics, C9 blind prediction, arrows, math-vs-physical falsifiers).	partial	C4 pre-asymptotics quantified (O8); C3 dictionary slot (O7); C9 pre-registration (O4); C1 remains register-held with its conditional evidence; arrows carry their own falsifiers (CMB phenomenology) and the free-sector impossibility is now a theorem; the math/physical falsifier split is partially implemented and remains open.
M10 (§4) Minor issues (cross-refs, $O(1)$ vs. exponentially small, floor, units, multi-index notation, bookmarks).	applied/already satisfied	Compile: 0 undefined references; defect restated; floor displayed; units paragraph; occupation multi-index convention; bookmarks verified.

4 Verification computations

4.1 Counting-law defect (O1)

At level m the defect extremum is $-\log(1+1/m)$, and $1/m = e^{-x}$ at $x = \log m$: amplitude ratio $\rightarrow 1$ (computed to $m = 10^6$ exactly and $m = 10^{100}$ by formula). The bound $-\log(1-e^{-x}) \leq 2e^{-x}$ holds at $x = \log 2$ with slack ($0.693 \leq 1.0$). **grok's non-convergence claim is false; muse's estimate is sharp.**

4.2 Conjugacy-class divisibility (O4)

Brute force over C_n ($n \leq 5$), S_n ($n \leq 5$), D_n ($n \leq 8$), Q_8 : every class size divides the group order, as orbit-stabilizer guarantees. **grok's "only in trivial cases" is wrong**; the paper's $c_i \mid N$ with $N = |G|$ is exact.

4.3 Dirichlet return-time bound (G9)

For m rationally independent frequencies, the pigeonhole grid with $N = \lceil 2\pi/\delta \rceil$ produces $q \leq N^m$ with all phases within $2\pi/N$: verified for $m = 2, 3$ at $\delta = 0.05$ and $m = 4, 5$ at coarser δ . This is the mechanism behind the new Proposition (return bound) $\tau_\varepsilon \leq (2\pi C_F/\varepsilon)^{m_F}$, $C_F = 2 + K_F m_F$.

4.4 Arithmetic vs. generic recurrence scales (G9/O1)

Simulated $\omega(n)$ near $x = 10^6, 10^{12}, 10^{18}$: means 2.94, 3.55, 3.76 vs. $\log \log x = 2.63, 3.32, 3.72$. With M integers near x , $m_F \lesssim M \log \log x$, hence $\tau \lesssim (C/\varepsilon)^{M \log \log x} = (S/k_B)^{O(1)}$: *polynomial in the entropy* for bounded support, against the generic-system scale $\gtrsim \varepsilon^{-D}$ with $D = e^{S/k_B}$: *double-exponential*. This is precisely the recurrence-time separation grok demanded as the only non-absorbable C2 falsifier.

5 New numerical controls (O8)

Platform	$\langle r \rangle$ arith	$\langle r \rangle$ scr	PR/ D arith	PR/ D scr	σ_{ETH} arith	σ_{ETH} scr
$d = 4, K = 11, V = 0$	0.581	0.510	0.238	0.113	0.36	0.37
$d = 4, K = 11, V = 0.3$	0.518	0.512	0.136	0.117	1.09	1.06
$d = 4, K = 11, \text{matched}$	0.509	0.501	0.138	0.117	1.08	1.08
$d = 3, K = 28, V = 0$	0.514	0.544	0.182	0.015	1.79	4.55

Reading: at matched kinetic dose the level statistics are scramble-insensitive (the GOE band is a property of the graph); the generic family’s delocalization is not (the arithmetic diagonal organizes transport through incommensurate step resonance). Generic-coupling quantile sweep at $d = 3, K = 28$: 0.464–0.540 (the earlier three-interior-quantile flatness was a sampling artifact; now superseded and reported as such). $1/\log D$ fits along the matched-dose ladders: pooled deviation $-1.66/\log D + 0.141$; residuals large; the limiting value is unsettled within the computed range.

6 Manuscript changes

1. **Theorem (log-count) strengthened:** sharp defect bounds, exponential decay, sharpness, fixed-window smoothing item, staircase honesty (§6).
2. **New Proposition (ensemble bridge):** $Z = \zeta$ as the exact Laplace–Stieltjes transform of the counting function; pole = transform of the exponential law; $\zeta - 1/(\sigma - 1)$ entire (§2).
3. **New Proposition (quantitative return times):** Dirichlet bound, polynomial-in-entropy arithmetic scale vs. double-exponential generic scale; C2 falsifier restated as a scaling criterion (§4).
4. **Ray periodicity classification** added to the no-period theorem (§4).
5. **Conjecture C3/Theorem (BH) restructured:** dictionary error slot $s(A) = O(\log(A/\ell_P^2))$; area coefficient exact; pure form falsifiable (§6, register).
6. **Clock section:** regulator ν disambiguated from inverse temperature; kinematical-level scoping; no-directional-limit addendum (§5).
7. **Dynamics section:** commutator identity $[H_P, M_m]$; provenance (“selected, not derived”); full generic quantile sweep; new Remark items (f) label-scrambled controls and (g) finite-size scaling; Figure (controls) (§3).
8. **Register:** C2, C3, C4, C5, C9, C10 rows updated; §1.2 operational-testability rule with interpretive exemption; falsifiability items 2 and 4 rewritten.
9. **Arithmetic sector:** incomplete-tensor-product/Fock unitary; explicit $D(H_P)$; spectrum qualifiers; Hagedorn structural difference; units paragraph; “not implied” boundary paragraph (§2).
10. **Galois section:** dictionary bookkeeping boundary (no gauge dynamics constructed) (§7).

Compile status: 45 pages (46 with cover), 0 overfull boxes, 0 undefined references; single-file manuscript generated and verified to compile identically.

7 Residual open items

1. C9 has no pre-registered application yet; the protocol’s well-formedness conditions are stated but unexercised on real data (all obtainable data admit no fit, which is compatibility, not confirmation).
2. The strong-ETH leg of C4 remains flat at all computed scales; no closing trend in $|\text{diag} - \text{micro}|/K$; the scaling limit of $\langle r \rangle$ is unsettled within the computed range.
3. muse’s demand for a strict separation of mathematical and physical falsifiers in §11 is only partially implemented.
4. grok’s editorial suggestion to split the manuscript into a mathematical note and a conjectural essay is recorded as a publishing option; not adopted.
5. The $H_{\text{tot}}^{(U,V)}$ family is selected, not derived; the infinite-volume self-adjointness of the kinetic completion (unbounded coefficients) is outside the Kato–Rellich argument and remains open.